

The Social Transmission of Prosociality from AI Agents: Examining the Effects of Observing AI Agents Lexical Alignment on Human-Human Interaction

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Abstract

Prosocial behaviors, such as empathy, are well-documented to be transmitted through observational learning. Linguistic alignment, the tendency of a more linguistically competent speaker to adapt to a less competent speaker, is considered a prosocial behavior as it supports others' emotional states and communication needs, so that alleviate others' negative states or unmet needs. While AI agents are increasingly designed to align linguistically with humans, it remains unclear whether their prosocial behaviors can influence human-human interactions. Specifically, can observing human-AI interactions impact how humans behave with one another? To address this, we conducted a preliminary experiment ($N = 40$) to examine how observing AI agents affect participants' prosocial linguistic behavior with children. We manipulated the agents' degree of lexical alignment (high vs. low) during communication. Results showed that observing a highly aligned AI agent but not low aligned AI agent increased participants' alignment with child interlocutor. Furthermore, highly aligned AI agents were perceived as more trustworthy, empathetic, and perspective-taking. These findings suggest that AI prosocial behavior can influence human prosociality and transfer to human-human interactions, with potential implications for designing AI systems to support human well-being.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Linguistic Alignment, Prosocial Behaviors, Human-AI interactions, Observational Learning, Conversational Design

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1 Introduction

Prosocial behaviors, such as empathy, represent a specific category of social behavior that is not only affiliative but also driven by the motivation to benefit others, resulting in improvements to their welfare [1]. Such behaviors can spread across individuals. Supporting this notion, there is evidence that self-reported prosocial behaviour increases if such behaviour is highly valued by others [2, 3], and when watching these behaviour of others [4]. Moreover, recent studies suggest that prosocial behaviour is socially transmitted through observational learning mechanisms [5]. The social transmission of prosociality is based on a learning process that is triggered by observing the prosocial behaviors of others. For example, observing empathic reactions to the pain of others affects learning of empathic responses in the observer: watching empathic or non-empathic responses generates a learning signal that respectively increases or decreases empathy ratings of the observer [6].

However, there has been a debate about the social transmission of prosocial behavior from technology to human behavior. For example, study showed that participants predicted the ratings of human demonstrators and computer demonstrators similarly well, the observations of the computer influenced the empathy ratings of participants to a lesser extent than the observations of the human demonstrator [6]. However, such study used computers as the demonstrator with little anthropomorphic design since they only presented the computer behavior to the participants in caption. Here we asked whether humans learn from a more anthropomorphic AI agent with human-like appearance, and human-like voice.

Linguistic alignment refers to the phenomenon where dialogue participants adjust their language use—such as vocabulary, syntax, and speech patterns—towards one another during interaction. This process serves as a means of establishing common ground and shared understanding, ultimately enhancing communicative effectiveness [7]. Researchers found that agents exhibit linguistic alignment towards their conversational partner and, as a result, the language of the two agent groups becomes more similar after interaction [8]. This finding suggests that LLMs can adapt their linguistic behavior during conversations, though the underlying mechanisms may differ from human alignment.

More important to the current research, linguistic alignment can be perceived as prosocial behavior, as it encapsulates various mechanisms that foster social bonds and cooperation among individuals. These mechanisms are articulated through frameworks such as goal-directed and social-affective mechanisms. When speakers linguistically align, they indicate a cooperative spirit and a

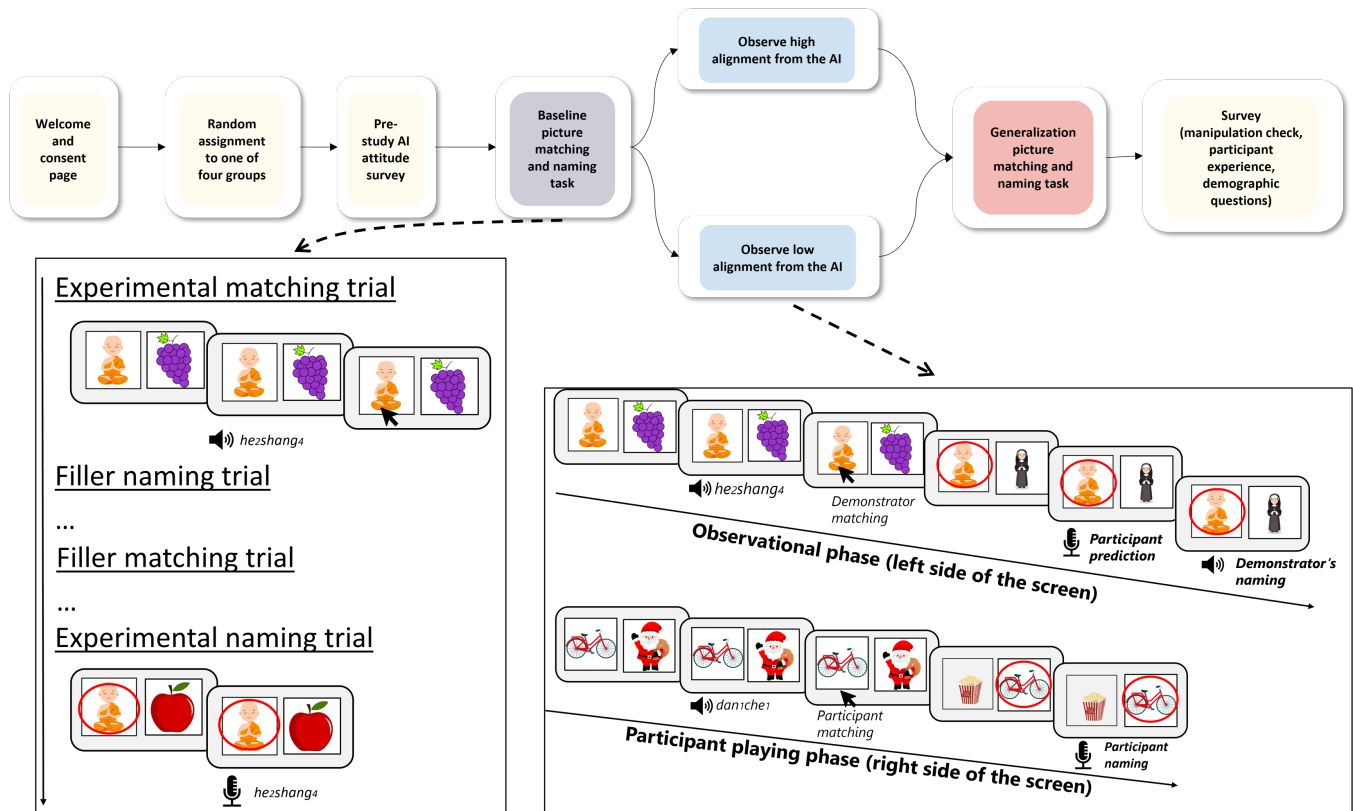


Figure 1: Experimental setup. The main experiment consisted of the baseline session, observational alignment learning session, and the generalization session. In each session, participants engaged in picture matching and naming games with simulated online partners, taking turns matching and naming target pictures. The interlocutors' naming preferences between preferred name and dispreferred name were balanced across conditions. During the observational learning session, participants observed the naming behavior of an AI agent demonstrator interacting with the same interlocutor. The demonstrator's responses were generated by AI text-to-speech synthesis based on a pre-defined script that determined alignment with the interlocutor's naming preferences according to the experimental group assignment (i.e., high alignment or low alignment group). Each trial consisted of an observation phase followed by a participant playing phase. In the observation phase, participants observed the interlocutor naming a target picture for the demonstrator, followed by the demonstrator's matching response. Participants were then shown two pictures and asked to predict the demonstrator's naming choice for the target picture (marked with a red circle). The demonstrator's actual naming was subsequently revealed. The participant playing phase began with an arrow indicating the switch to direct interaction. Participants first matched a picture based on the interlocutor's description, then named a target picture (marked with a red circle) for the interlocutor to match. Variable delays (3-5 seconds) simulated realistic interaction timing throughout both phases.

desire to work toward mutual goals. Research has shown that effective communication relies significantly on this alignment, as it not only strengthens interactions but also enhances the likelihood of successful exchanges of ideas and emotions [7, 9]. Interactional alignment simplifies the process of language production and comprehension, allowing for more fluid and effective communication [7]. Furthermore, audience design theory posits that speakers adapt their language to suit the perceived needs and expectations of their listeners, thereby enhancing mutual understanding [10]. This form of alignment can be inherently prosocial as it signifies an awareness and consideration of the audience's background, preferences, and perspectives. Therefore, it is crucial to examine how learning

from AI's linguistic alignment in communication affects humans' linguistic alignment.

In this study, we examine how observing AI agents influences human prosociality and, specifically, how learning from AI's high or low linguistic alignment in communication affects humans' linguistic alignment to the same interlocutor and to other interlocutors. We conducted an offline experiment ($N = 40$) with two main conditions: one where participants observed a highly aligned AI agent, and another where they observed a low aligned AI agent. Within each group, we further assigned participants into two groups where the child interlocutor either uses preferred name or dispreferred

name to name a picture. If participants align with the child interlocutor, there should be more dispreferred responses than preferred responses when the child interlocutor uses dispreferred name, and vice versa. For each group, we measured participant's alignment effect with two different child interlocutors before and after their observation of the AI agent's interaction with a child interlocutor. Our findings showed that observing a highly aligned AI agent, but not low aligned AI agent increased participants' alignment with children. Furthermore, highly aligned AI agents were perceived as more trustworthy, empathetic, and perspective taking.

2 Methods

The experiment structure adopted an observational learning paradigm similar to that used in [6] and the tasks adopted a picture matching and naming task similar to [11]. Participants were informed before the experiment that they were to play a picture matching and naming game with other players on the internet. The experiment had a design of 2 (observational group: high vs. low) $\times$ 2 (prime: preferred name vs. dispreferred name), with both variables manipulated between participants and within items.

2.1 Participants

We recruited 40 (32 female and 8 male) participants in this study with an age range between 18 and 36 ($M = 25.6$). All participants were native Mandarin speakers, and all were university or graduate students.

2.2 Procedure

The experiment was conducted offline using PsychoPy [12]. Participants were instructed to wear a headset with a microphone. After providing informed consent, participants were shown an on-screen instruction in Chinese. As shown in Figure 1, participants engaged in simulated online picture matching and naming games using pre-recorded interlocutors. The experiment comprised three sessions: baseline, observational alignment learning, and generalization. In baseline and generalization sessions, participants alternated between matching pictures based on child's references and naming target pictures for their "partner". The key observational learning session used a two-phase design where participants first observed a demonstrator's interactions with the interlocutor, then played themselves. Participants were assigned to either high alignment (demonstrator aligned with interlocutor's naming preferences in 77.8% of trials) or low alignment conditions (22.2% alignment). The demonstrator's responses were AI-synthesized and systematically varied to manipulate observed alignment behavior. Each session included multiple trials with matching and naming components. The observational session contained 36 trials across two blocks with different interlocutors to prevent habituation. Participants completed 72 runs total across approximately 60 minutes.

3 Results and discussion

3.1 Manipulation check

We conducted a manipulation check to confirm the effectiveness of our manipulation regarding the AI demonstrator's alignment behavior. Specifically, we used a yes/no item to assess participants'

perception of alignment: "AI agents in the second game repeated the references of the child player when naming the same picture." Results indicated that participants effectively perceived the variation in alignment rates. Among participants assigned to the high alignment condition, 89% believed their AI demonstrator mostly aligned with the child interlocutor, while 92% of those in the low alignment condition believed their AI demonstrator seldom aligned with the child interlocutor.

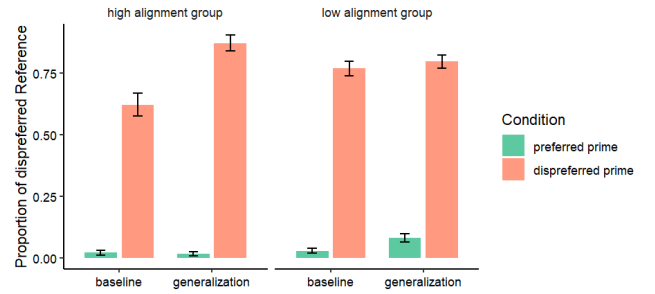


Figure 2: Proportion of dispreferred responses as a function of prime type, group type and session. (error bars representing $\pm$ SE)

3.2 Observing human-AI alignment affects human-human alignment

Following [11] and [13], the lexical alignment effect was calculated as the proportion of producing dispreferred name after dispreferred name prime minus the proportion of producing dispreferred name after preferred name prime.

We next subjected all responses to a logit mixed effects (LME) analysis, using Prime (preferred vs dispreferred), Group (high vs low) and Session (baseline vs generalization) as interacting predictors. Our model comparison procedure supported a random effects structure that included by-participant random slopes for Session and by-item random slopes for the full three-way interaction between Prime, Group, and Session.

The analysis revealed a significant main effect of Prime ($\beta = 8.49$, $SE = 1.19$, $z = 7.13$, $p < .001$), indicating that participants were more likely to produce dispreferred responses following dispreferred primes compared to preferred primes. There was no significant main effect of Group ($\beta = -0.71$, $SE = 1.09$, $z = -0.66$, $p = .51$) or Session ($\beta = 1.04$, $SE = 1.03$, $z = 1.01$, $p = .31$). However, we observed a significant three-way interaction between Prime, Group, and Session ($\beta = 8.43$, $SE = 4.03$, $z = 2.09$, $p = .036$), suggesting that the effect of Prime varied across Group and Session conditions.

To further investigate this three-way interaction, we conducted separate analyses for the high and low alignment groups. For the high alignment group, we found significant main effects of Prime ($\beta = 7.24$, $SE = 1.08$, $z = 6.71$, $p < .001$) and Session ($\beta = 1.15$, $SE = 0.51$, $z = 2.26$, $p = .024$), as well as a significant Prime $\times$ Session interaction ($\beta = 2.71$, $SE = 1.01$, $z = 2.70$, $p = .007$). As shown in Figure 2, this interaction indicates that the effect of Prime was stronger in the generalization session than in the

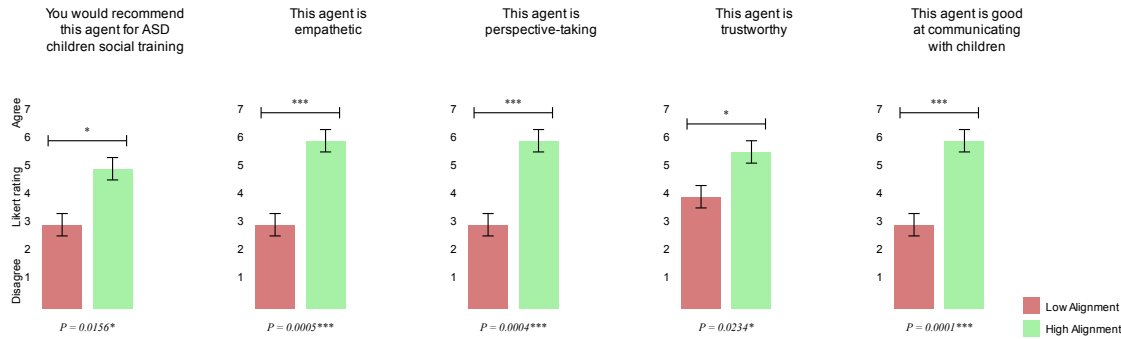


Figure 3: Results of participant ratings on Likert scales relating to trust, empathy and perceived effectiveness of social interaction. ASD stands for Autism Syndrome Disorder.

baseline session for the high alignment group, suggesting enhanced alignment effects after observing high alignment AI agent.

For the low alignment group, we found a significant main effect of Prime ($\beta = 6.55$, $SE = 0.75$, $z = 8.75$, $p < .001$), but no significant effect of Session ($\beta = 0.80$, $SE = 0.54$, $z = 1.49$, $p = .137$) or Prime $\times$ Session interaction ($\beta = -0.61$, $SE = 1.04$, $z = -0.58$, $p = .559$). This indicates that while the low alignment group showed strong priming effects, these effects did not significantly change from baseline to generalization sessions.

These results suggest that participants in the high alignment group demonstrated greater adaptability in their linguistic choices after observing AI agent highly aligning with the child interlocutor, while the low alignment group maintained relatively stable response patterns regardless of session.

3.3 Alignment pattern can alter experiences and perceptions of AI

Observing AI agents' alignment behavior significantly influenced participants' perceptions and evaluations of the agents across multiple dimensions. As shown in Figure 3, participants who observed highly aligned AI agents consistently rated them more favorably than those who observed low alignment agents across all measured constructs.

Specifically, participants in the high alignment condition rated the AI agent as significantly more empathetic ($M = 5.95$, $SD = 0.83$) compared to the low alignment condition ($M = 3.05$, $SD = 1.43$, $p = 0.0005$). They also perceived the agent as demonstrating greater perspective-taking ($M = 5.90$, $SD = 0.85$) than participants in the low alignment condition ($M = 3.05$, $SD = 1.43$, $p = 0.0004$). Additionally, the high alignment group provided significantly higher trustworthiness ratings ($M = 5.55$, $SD = 0.83$) compared to the low alignment group ($M = 4.05$, $SD = 1.32$, $p = 0.0234$).

Regarding practical applications, participants assigned significantly higher ratings to the statement "you would recommend this agent for ASD children social training" when they observed the high alignment agent ($M = 4.95$, $SD = 1.50$) compared to the low

alignment agent ($M = 3.05$, $SD = 1.43$, $p = 0.0156$). Similarly, participants rated the high alignment agent as significantly more effective at communicating with children ($M = 5.95$, $SD = 0.89$) compared to the low alignment agent ($M = 3.05$, $SD = 1.36$, $p = 0.0001$).

These findings demonstrate that AI agents' alignment behavior serves as a powerful signal for participants' social and functional evaluations. The consistently higher ratings across empathy, perspective-taking, trustworthiness, and communication effectiveness suggest that linguistic alignment functions as a key indicator of prosocial competence, influencing how humans perceive AI agents' social and therapeutic potential.

4 Limitation and future work

Several limitations should be acknowledged in our findings. Our study involved a small, homogeneous sample ($N = 40$, Chinese university students), limiting generalizability across diverse populations and cultures. We only measured immediate short-term effects, leaving questions about the persistence of alignment behaviors over time. Future research should address these limitations through several directions. Longitudinal studies are needed to examine whether alignment effects persist over days or weeks and whether repeated exposure leads to lasting behavioral change. Field deployments with actual AI systems in naturalistic ubiquitous computing environments (smart homes, mobile assistants) would provide greater ecological validity.

5 Conclusion

Our research presents several contributions to HCI. Firstly, we offer empirical evidence demonstrating that observing the prosocial behaviors of AI agents can elicit human prosociality. Thus, it supports the new sociology of human and machine hypothesis [14] that observing M-M or H-M interactions can influence how humans act towards each other. This suggests that, amidst the growing trend of AI helping people, it is important to go beyond just focusing on improving AI performance but also consider enhancing human prosociality during these interactions. Secondly, we offer insight

into whether AI with high linguistic alignment could be perceived as empathetic, perspective-taking and trustworthy behavior. This offers guidance on designing AI to better support human and specifically children communication and enhance human prosociality in scenarios where AI acts as a demonstrator. Thirdly, we offer a novel perspective on AI-aided intervention to improve perspective-taking and empathy for children with Autistic Spectrum Disorders, as our findings suggest that in addition to use personal AI agent to help these children communicating with other typical developing children, we can also engage AI demonstrator to help ASD children to communicate with other children themselves.

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